

Problem-Solution Fit

Date	02 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being (HDI Prediction System)
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience.

1. CUSTOMER SEGMENT(S) [CS] • Policymakers: Government officials planning national budgets and development goals. • Researchers: Academics studying global economic and social disparities. • Students/Analysts: University students and data analysts working on socio-economic projects.	6. CUSTOMER LIMITATIONS [CL] • Technical Skills: Policymakers and students often lack advanced programming skills (Python/R) to process raw data. • Budget: Limited funds for expensive, enterprise-grade Business Intelligence (BI) or statistical software. • Devices/Connectivity: Reliance on standard laptops; users in developing regions may have intermittent internet access.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Current Solutions: UNDP Static Reports, Manual Excel Calculations, SPSS/R. Pros: Highly accurate, officially recognized, comprehensive historical data. Cons: Completely static (no real-time "what-if" analysis), steep learning curve for statistical software, highly time-consuming to update and model manually.
2. PROBLEMS / PAINS + FREQUENCY [PR] • Pain: Inability to instantly simulate how changes in specific sectors (e.g., increasing education funding) will shift a country's overall HDI tier. • Pain: Manual calculation	9. PROBLEM ROOT / CAUSE [RC] The core issue is the lack of accessible, interactive, and real-time predictive tools that translate raw, multi-dimensional socio-economic data into actionable HDI insights without requiring advanced	7. BEHAVIOR + INTENSITY [BE] • Behavior: Users manually download massive CSV datasets from the World Bank/UNDP, spend hours cleaning data in Excel, manually apply complex HDI formulas, and create

of the HDI geometric mean is tedious and prone to human error. • Frequency: High – Occurs frequently during national budget planning, policy drafting, and academic research cycles.	data science or programming expertise.	static charts. • Intensity: High effort, high time investment, low agility. Users are stuck in a repetitive, manual data-processing loop.
3. TRIGGERS TO ACT [TR]	10. YOUR SOLUTION [SL]	8. CHANNELS OF BEHAVIOR [CH]
• Need to draft a new national development policy or justify budget allocations. • Requirement to publish a research paper or complete a university data analysis project. • Release of new annual UNDP Human Development Reports prompting updated comparative analysis.	An interactive, web-based Machine Learning application (built with Flask & Scikit-learn) that allows users to input or upload country indicators (Life Expectancy, Education, GNI) to instantly predict the HDI score and tier (Very High to Low), complete with intuitive data visualizations.	ONLINE: • World Bank / UNDP Open Data portals (downloading raw data). • Web browsers accessing our Flask Web App. • Excel / SPSS for manual processing. OFFLINE: • Policy planning meetings and government boardrooms. • Academic classrooms and research presentations.
4. EMOTIONS BEFORE / AFTER [EM]		
Before: Overwhelmed by complex datasets, anxious about making incorrect policy decisions based on outdated static reports, frustrated by the time-consuming nature of manual calculations. After: Confident in data-driven decision-making, relieved by automated and instant calculations, empowered by interactive visualizations and scenario modeling.		